

Two Interfaces, Twelve Specialists

AP Statistics · Unit 4 · Topic 4.2 · B: 2027-01-28 · E: 2027-01-29 · TEACHER KEY

CED TETHER

- **UNC-4.O / EK O.1, O.3 [Skill 1.D]** Use a one-sample t interval for a mean with unknown population SD. Matched-pair differences form one sample.
- **UNC-4.P / EK P.1 [Skill 4.C]** Check random sampling or assignment, the 10 percent condition, and shape on the differences.
- **UNC-4.Q / EK Q.1–Q.3 [Skill 3.D]** With $df = n - 1$, $SE = s_d/\sqrt{n}$ and margin of error is t^*SE .
- **UNC-4.R / EK R.1–R.2 [Skill 3.D]** Use $\bar{d}$ as the estimate and calculate $\bar{d} \pm t^*SE$ for the population mean difference.

Essential question: How do linked measurements choose an interval and bound a causal claim?

DOK-3 skill: 4.C · **FRQ pattern:** paired-vs-independent-interval-and-scope

DOK rationale: Part (c) selects a procedure from the pairing, uses interval and SRS scope, diagnoses fixed-order confounding, and traces the independence and causal consequences.

First take → rules + (a)–(c) → turn in

DOK: 1 → 3

Minutes: 45

Teacher does

- Display the board and ask for one sentence using the study description.
- At minute 5, read the rules box aloud once; students start (a).
- Return to the board at minute 33. Students finish the shortened parts and justify their judgment.
- Collect at the door. Score part (c) only using the three required elements.

Students do

- Write a one-sentence first take before any discussion.
- Work (a) and (b) from the rules box and the stem.
- Finish (a)–(c), revise the first take in the exit line, and turn in the sheet.

Questions to ask

- Which specific number or study detail supports your choice?
- What claim would go beyond the evidence, and why?

Adult role

Circulate and ask for evidence. Accept a concise response; do not require omitted calculations or a second full inference write-up.

[WARN] LOOK FOR

- Choosing a speaker without a reason.
- Copying numbers without explaining what they support.

SCORING PART (c) — E / P / I

Expected elements

- **preserves-pairs** (required) — Chooses Priya and explains the within-person link
- **interprets-interval** (required) — Interprets the positive interval for the 180 specialists under this sequence
- **bounds-cause** (required) — Explains a practice/time alternative from fixed order and no comparison group

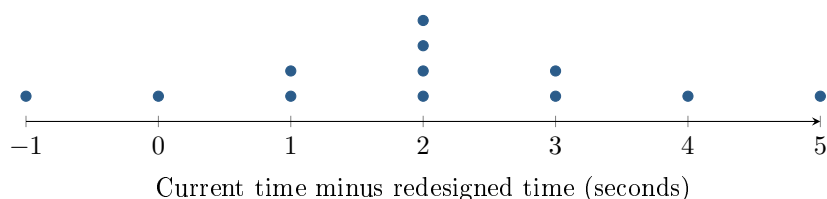
E	Justifies paired analysis, interprets the mean interval in context, and explains the practice/time limit on causation.
P	Shows substantive valid reasoning for at least one required element, but does not meet all E requirements. Credit correct reasoning even when another claim is wrong.
I	Shows no substantive valid reasoning for any required element; a name, unsupported choice, or copied number alone does not count.

[STAR] TODAY'S PROBLEM — annotated

Suppose a hospital randomly samples 12 of its 180 billing specialists. Each does the same task on the current computer screen and, one week later, on a redesigned screen. Everyone uses the current screen first; there is no comparison group. Let d = current – redesigned time in seconds. The plot shows the differences; $\bar{d} = 2.000$, $s_d = 1.6514$, and the 95% critical value is $t^* = 2.201$.

Priya: “Use one sample of paired differences. Doing the tasks in a fixed order limits a cause-and-effect claim.”

Mateo: “Two columns require two independent samples. A positive interval proves the redesign caused faster work.”



FIRST TAKE (5 min)

Which seems more useful here: each person's change in time or two separate groups of times? Explain in one sentence.

Key: Ungraded. Everyday language is enough before the discussion; formal vocabulary belongs in the finish.

Rules callout “ONE-SAMPLE MEAN INTERVALS (keep on the page)” is printed on the student sheet.

DOK 1 (a) How many paired differences are there, and what does a positive difference mean?

Key: There are 12 paired differences. A positive current-minus-redesigned difference means the later task took less time.

DOK 2 (b) Use the supplied summaries to calculate the 95% interval $\bar{d} \pm t^*(s_d/\sqrt{n})$. Show one calculation.

Key: $2.000 \pm 2.201(1.6514/\sqrt{12}) = 2.000 \pm 1.0493 = (0.9507, 3.0493)$ seconds (small rounding differences accepted).

DOK 3 (c) In 3–4 sentences, choose an analysis and justify it from the linked measurements. Interpret the interval for the 180 specialists, then explain why practice or the fixed order limits a causal claim.

Key: Priya is better supported because each specialist supplies two linked measurements, so the 12 differences form one sample. Treating the columns as independent loses that link and changes the standard error. We are 95% confident the mean current-minus-redesigned time for the 180 specialists under this sequence is about 0.95 to 3.05 seconds. Because the current screen always came first and there was no comparison group, practice or time could explain the improvement; the redesign alone is not shown to cause it.

EXIT

One thing the rules box changed about my first take: _____